3 Heavy metal contamination in water poses significant risks to environmental and human well-being. Common heavy metals found in water include cadmium (Cd), lead (Pb) and mercury (Hg).

(a) The standard electrode potential of the $Cd^{2+}(aq)/Cd(s)$ electrode is −0.403V.

(i) Define the term *standard electrode potential, $E^{\ominus}$*.

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..... **[1]**

(ii) Draw a fully labelled diagram of the experimental set-up used to measure the standard electrode potential, $E^{\ominus}$, of the $Cd^{2+}(aq)/Cd(s)$ half-cell.

[2]

(iii) Predict how the electrode potential, $E^{\ominus}$, of $Cd^{2+}(aq)/Cd(s)$ will be affected when aqueous sodium hydroxide is added to the $Cd^{2+}(aq)/Cd(s)$ half-cell. Explain your answer.

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..... **[2]**

(b) A water source was found to be contaminated by heavy metal contaminant, Pb^{2+}.

To determine whether the water is safe to drink, complexometric back titration was employed to determine the concentration of Pb^{2+} in a water sample.

The water sample containing Pb^{2+} will be reacted with an excess amount of $EDTA^{4-}$, where a lead-EDTA complex will be formed in the process.

The general reaction of metal ion, M^{2+} and $EDTA^{4-}$ is as shown:

$$M^{2+} + EDTA^{4-} \rightleftharpoons [M(EDTA)]^{2-}$$

The remaining amount of $EDTA^{4-}$ is then determined by titrating with zinc sulfate, with Eriochrome Black T as an indicator.

(i) 10.0 cm^3 of 5.0 x 10^{-7} mol dm^{-3} of $EDTA^{4-}$ was added to 10.0 cm^3 of water sample containing Pb^{2+}. The resulting solution was found to require 10.0 cm^3 of 2.0 x 10^{-7} mol dm^{-3} of zinc sulfate solution for complete reaction.

Calculate the amount, in moles, of Pb^{2+} present in 10.0 cm^3 of the water sample.

[1]

(ii) Calculate the mass of Pb^{2+}, in mg, present in 1 dm^3 of water sample.

Given that the safe limit of maximum mass of Pb^{2+} is 0.0100 mg dm^{-3}, comment on whether the water is safe to drink.

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..... **[2]**

(iii) Hydrogen sulfide, H_2S, is added to another 1 dm^3 of water sample containing 1.0 x 10^{-9} mol dm^{-3} of Hg^{2+} and Pb^{2+}each.

Table 3.1 shows the K_{sp} values for the corresponding metal sulfides.

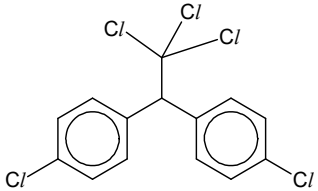
Table 3.1

Metal Sulfide	K_{sp} / mol^2dm^{-6}
PbS	9 x 10^{-29}
HgS	2 x 10^{-53}

Calculate the minimum concentration of hydrogen sulfide added to remove the maximum concentration of Hg^{2+} without precipitating Pb^{2+}. Hence, determine the maximum mass of HgS precipitated in 1 dm^3.

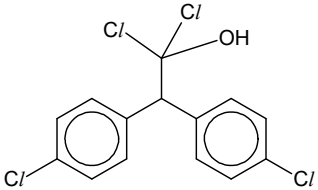
[2]

(c) DDT is a common ingredient in insecticides and it can enter groundwater as an organic pollutant through processes like runoff and leaching.



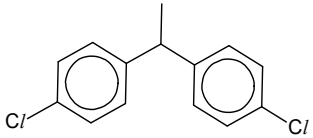
DDT

(i) A student proposed that compound **J** will be formed when DDT is reacted with hot aqueous sodium hydroxide, assuming that the rate of reaction is independent of the concentration of NaOH.



Compound J

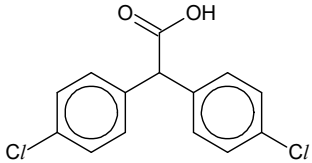
Name and describe the mechanism for the reaction between DDT and hot aqueous sodium hydroxide to form compound **J**. Include all relevant lone pairs, dipoles, curly arrows and charges. Include the structure of the organic intermediate.

You may represent 

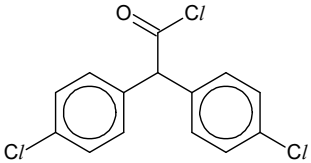
as R.

[2]

(ii) It was found that compound **K** is formed when DDT reacts with hot aqueous sodium hydroxide. **K** then reacts with phosphorus pentachloride to form compound **L**.



compound **K**



compound **L**

When the same amount of compounds **K** and **L** (not necessarily in that order) are added to separate and equal volumes of water, solutions are formed with pH values of 0.5 and 3.0.

Suggest which pH value is associated with compounds **K** and **L**. Explain your answer.

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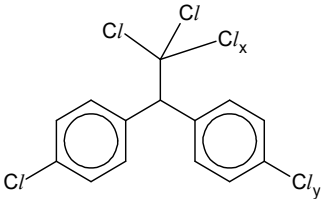
..... **[2]**

(iii) Explain the difference in reactivity of the two chlorine atoms labelled Cl_x and Cl_y in DDT towards hot aqueous sodium hydroxide.



DDT

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..... **[2]**

[Total: 16]